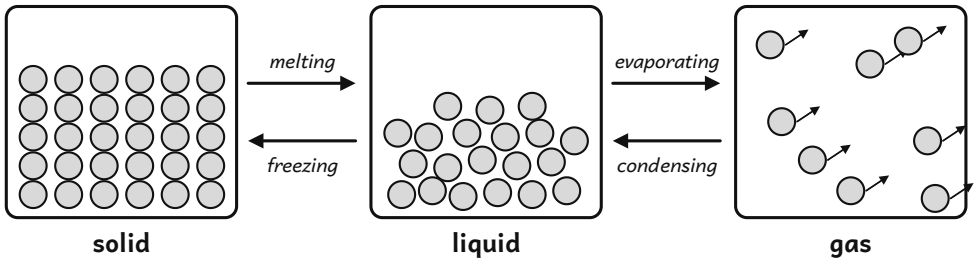
PHYSICAL AND CHEMICAL SCIENCE

States of Matter: Evaporation, Condensation, Boiling and Melting

Solids, liquids and gases, and how heating and cooling change them



In this pack

📖 **Learn it:** Solids, liquids and gases: changing state ★☆☆
Fact check: Changes of state ★☆☆
Think it through: Explaining changes of state ★★☆
Investigate: Does surface area affect evaporation? ★★☆
Real-world case study: Salt from the sun: Cox's Bazar salt farmers ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

© InquiryMind · Learn · Play · Wonder · Free resources for everyone, bringing quality and equity to education and helping teachers build concepts and make learning visible · Illustrations drawn for InquiryMind · Print in black and white on A4 · Challenge: ★ core ★★ deeper ★★★ stretch

Solids, liquids and gases: changing state

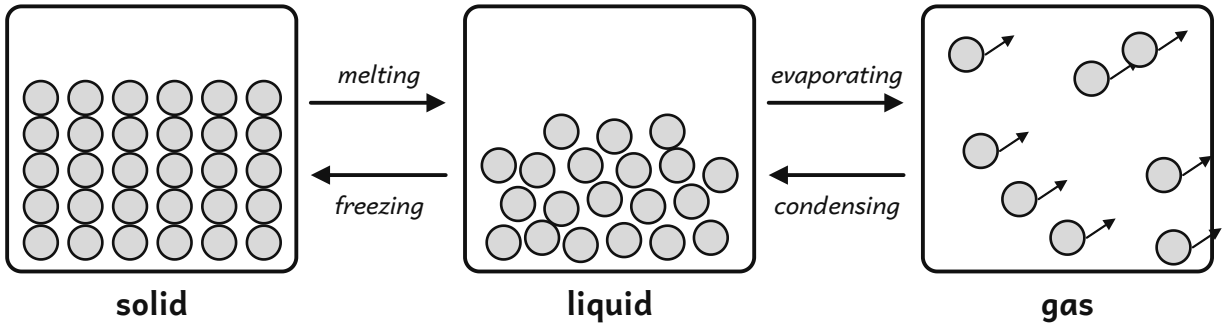
★☆☆ Grade 5

Name: Class: Date:

Learning goals

- Explain evaporation, condensation, boiling, melting and freezing.
- Recall that water boils at 100 °C and ice melts at 0 °C.
- Explain that adding or removing heat causes changes of state.

Wet clothes dry on the roof. A glass of cold water "sweats" on a hot day. Ice in a glass of lemon sherbet melts. A kettle whistles as water boils. All of these are **changes of state**, and every one is caused by **heating** or **cooling**.



The particles in a solid, a liquid and a gas.

Three states of matter

In a **solid**, particles are packed tightly in rows and can only vibrate, so a solid keeps its shape. In a **liquid**, particles are close together but can move past each other, so a liquid flows and takes the shape of its container. In a **gas**, particles are far apart and move quickly in all directions, so a gas spreads out to fill any space.

Evaporation

Evaporation is when a liquid turns into a gas. It happens at the **surface** of a liquid, at any temperature. Water evaporates faster when it is **warmer**, when it is **windy**, and when it has a **larger surface area**. That is why clothes dry fastest when they are spread out on a hot, windy day.

Condensation

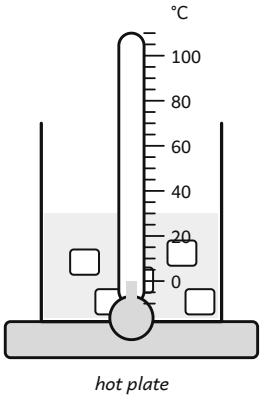
Condensation is the **reverse** of evaporation: a gas turns into a liquid when it **cools**. Air always contains some invisible **water vapour**. When it touches a cold surface, such as a cold glass, a window in winter or grass at night, it condenses into drops. This is how **dew** forms on winter mornings.

Boiling

When water is heated strongly it **boils** at **100 °C**. Bubbles of water vapour form all through the liquid, not just at the surface. While it is boiling, the temperature stays at 100 °C: the heat energy is being used to turn liquid into gas.

Melting and freezing

Ice **melts** at **0 °C**, and water **freezes** at 0 °C. Different materials melt at different temperatures: chocolate melts at about 30–35 °C (in your hand!), candle wax at about 60 °C, and iron at over 1500 °C.



Heating ice and recording the temperature.

Evaporating a solution

If you dissolve salt in water and then let the water **evaporate**, the salt is left behind as crystals. The salt does not evaporate. This is how salt is made from sea water.

From the past

The **Celsius** scale was invented in 1742 by the Swedish astronomer Anders Celsius. Surprisingly, he first set 0 for boiling water and 100 for freezing water! It was turned the other way round after he died. In the USA people still often use the **Fahrenheit** scale, on which water freezes at 32 °F and boils at 212 °F.

Did you know?

On top of Mount Everest water boils at only about 70 °C, because the air pressure is much lower. It is hard to cook rice up there!

Key words

evaporation	liquid → gas, at the surface, at any temperature
condensation	gas → liquid when it cools
boiling	liquid → gas all through the liquid, at the boiling point
melting	solid → liquid
freezing	liquid → solid
water vapour	water as a gas (invisible)
melting point	the temperature at which a solid melts
boiling point	the temperature at which a liquid boils

FACT CHECK · C1 · STATES OF MATTER

Changes of state

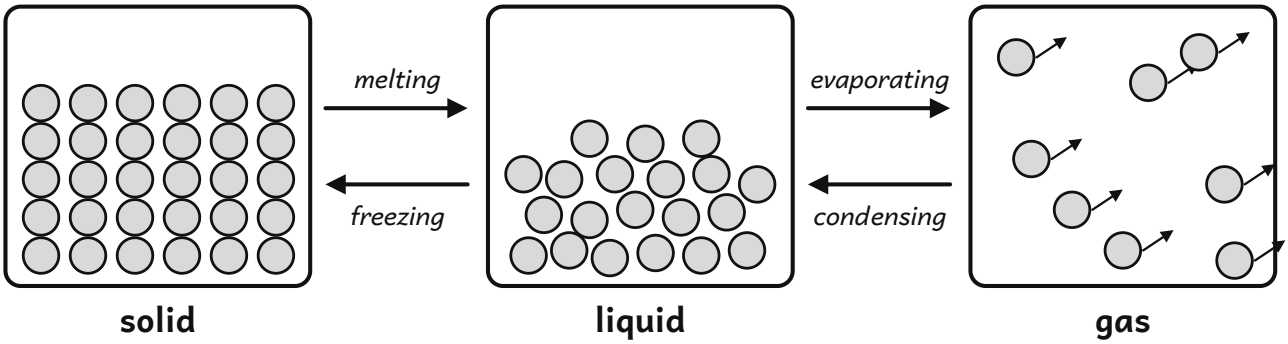
★☆☆
Grade 5

Name: Class: Date:

Learning goals

- Name the changes of state.
- Recall key temperatures.

1 **Label the changes of state on the arrows.** [4]



Change	Name
solid → liquid	
liquid → solid	
liquid → gas	
gas → liquid	

2 **Fill in the gaps.** [5]

WORD BANK

100 **50** **evaporates** **condenses** **salt** **melts** **0**

Water boils at _____ °C. Ice melts at _____ °C. When water vapour touches a cold window it _____. Puddles dry up because the water _____. When salty water evaporates, the _____ is left behind.

3 **Match each everyday event to the change of state.** [5]

Draw a line to match each pair, or write the letter.

wet clothes drying ●	● **A** melting
dew on the grass in the morning ●	● **B** boiling
ice cream dripping ●	● **C** freezing
a kettle making steam ●	● **D** condensation
water turning into ice cubes in a freezer ●	● **E** evaporation

4 **True or false?** [3]

Statement	True	False	If false, correct it
1. Evaporation only happens at 100 °C.			
2. Condensation is the reverse of evaporation.			
3. The temperature of boiling water keeps rising while it boils.			

THINK IT THROUGH · C1 · STATES OF MATTER

Explaining changes of state

★★☆
Grade 5

Name: Class: Date:

Learning goals

- Explain changes using heating and cooling.
- Interpret a heating graph.

1 **Explain each observation.** [6]

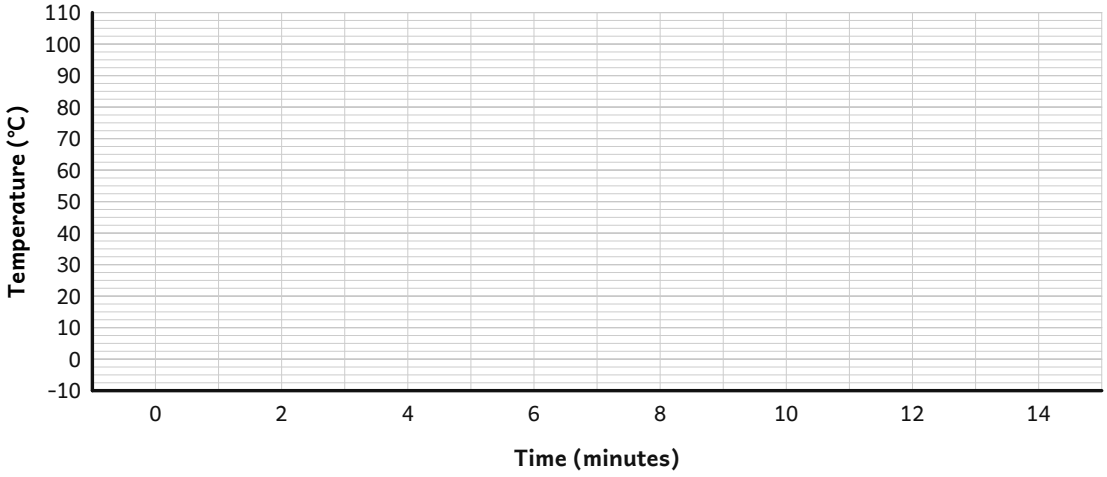
a) A cold bottle of water from the fridge becomes wet on the outside.

b) Clothes dry faster when they are spread out than when they are folded.

c) In winter, people's breath looks like a little cloud.

2 **Class 5A heated ice and recorded the temperature every minute.** [6]

Time (min)	0	2	4	6	8	10	12	14
Temp (°C)	−10	0	0	25	58	90	100	100



a) Plot the line graph.

b) Why did the temperature stay at 0 °C between 2 and 4 minutes?

.....

.....

c) What was happening at 12–14 minutes?

.....

.....

3 **Why do people in Bangladesh store drinking water in clay pots (*kolshi*) to keep it cool?** [2]

.....

.....

.....

INVESTIGATE · C1 · STATES OF MATTER

Does surface area affect evaporation?

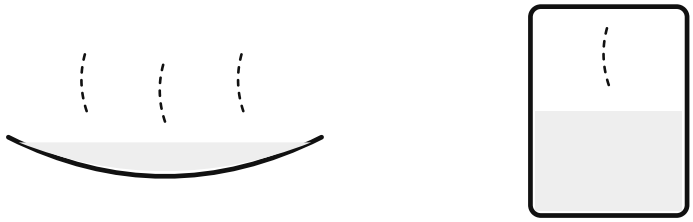
★★☆
Grade 5

Name: **Class:** **Date:**

Learning goals

- Plan a fair test.
- Plot two lines on one graph.
- Draw a conclusion.

same volume of water: 100 ml



wide dish **narrow cup**

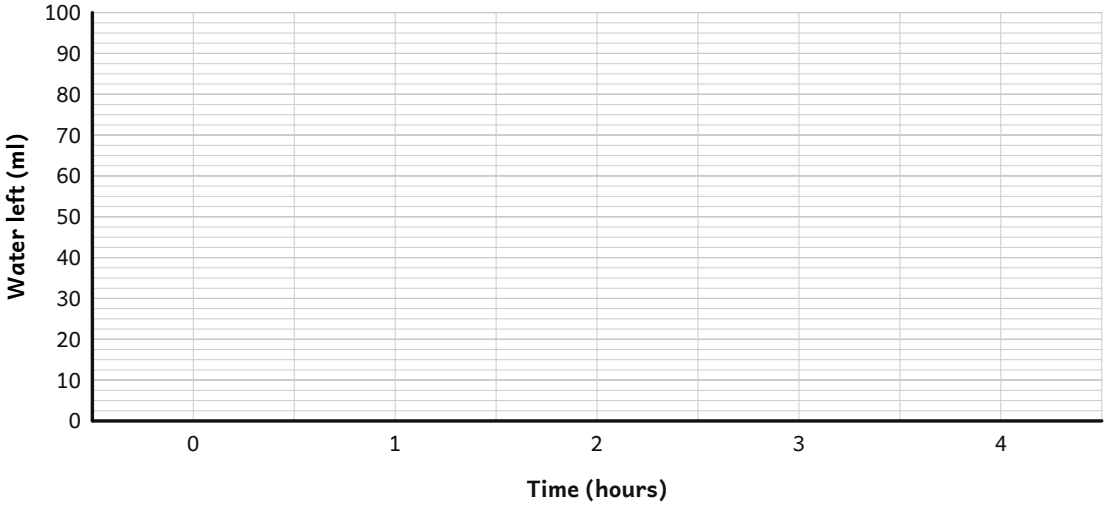
Arif and Mitu put 100 ml of water in a wide dish and 100 ml in a narrow cup. They left both in the same sunny place and measured the water left every hour.

1 **Complete the plan.** [5]

Question	
Change	
Measure	
Keep the same	
Prediction	

2 **Plot both sets of results on one line graph.** [5]

Time (hours)	0	1	2	3	4
Wide dish (ml)	100	82	64	46	30
Narrow cup (ml)	100	96	92	89	85



3 **How much water evaporated from each container in 4 hours?** [2]

4 **Write a conclusion.** [2]

Salt from the sun: Cox's Bazar salt farmers

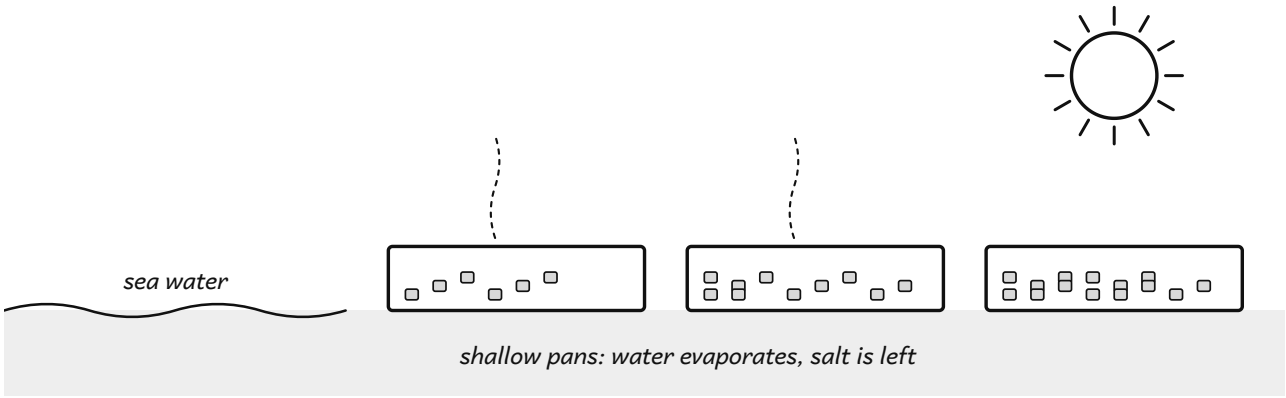
★★★ Grade 5

Name: Class: Date:

Learning goals

- Apply evaporation to a real industry.
- Use data and maths.
- Consider how weather and climate affect people.

Science | Geography | Maths | Economics | Health



Salt pans: sea water is let into shallow fields and the Sun evaporates the water.

Almost all the salt eaten in Bangladesh is made on the coast of **Cox's Bazar** and Chattogram. From about November to May, the dry season, farmers let sea water into shallow, flat fields covered with plastic sheets. The Sun and the wind **evaporate** the water, and white **salt crystals** are left behind. Farmers rake up the salt and carry it away in baskets.

In the monsoon, rain stops salt making: rain dissolves the salt and adds fresh water to the fields. Many salt farmers then grow shrimp or rice instead. Before it is sold, salt is cleaned and **iodine** is added, because our bodies need a small amount of iodine for healthy growth and brain development.

Based on descriptions of the Bangladesh salt industry (BSCIC and news reports). Numbers below are examples.

1 **Explain how salt is made using the words *evaporate*, *dissolved* and *crystals*.** [3]

2 **Why can salt not be made during the monsoon?** [2]

3 **About 3.5 kg of salt is dissolved in every 100 litres of sea water.** [3]

a) How much salt is in 1000 litres of sea water?

b) A farmer's field holds 20 000 litres. How much salt could it make?

4 **Why is iodine added to salt?** [1]

5 **Rising sea levels and stronger storms may affect salt farms. Suggest one problem and one solution.** [2]

Show what you know

★★★
Grade 5

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

J	W	S	R	B	D	J	C	J	H
G	A	S	T	V	A	P	O	U	R
M	F	S	M	E	V	S	N	M	E
E	Y	O	Y	V	A	K	D	Y	B
L	F	L	L	H	J	M	E	M	H
T	R	I	P	I	H	G	N	N	E
U	E	D	F	P	Q	O	S	P	G
I	E	A	G	K	B	U	E	H	J
R	Z	I	B	M	E	I	I	T	W
V	E	W	U	B	O	I	L	D	L

boil condense freeze gas liquid melt solid steam vapour

2 **At what temperature does pure water boil?** [1]

- ◯ 0 °C
- ◯ 37 °C
- ◯ 100 °C
- ◯ 212 °C

3 **Which change happens when dew forms?** [1]

- ◯ evaporation
- ◯ condensation
- ◯ melting
- ◯ boiling

4 **Draw the particles in *a* solid, *a* liquid *and a* gas.** [3]

solid

liquid

gas

5 **Challenge. Explain the difference between evaporation and boiling.** [2]

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
describe solids, liquids and gases using particles	○	○	○
explain evaporation and condensation	○	○	○
recall that water boils at 100 °C and ice melts at 0 °C	○	○	○
explain why the temperature stays the same while ice melts	○	○	○
investigate what affects evaporation	○	○	○
explain how salt is made from sea water	○	○	○